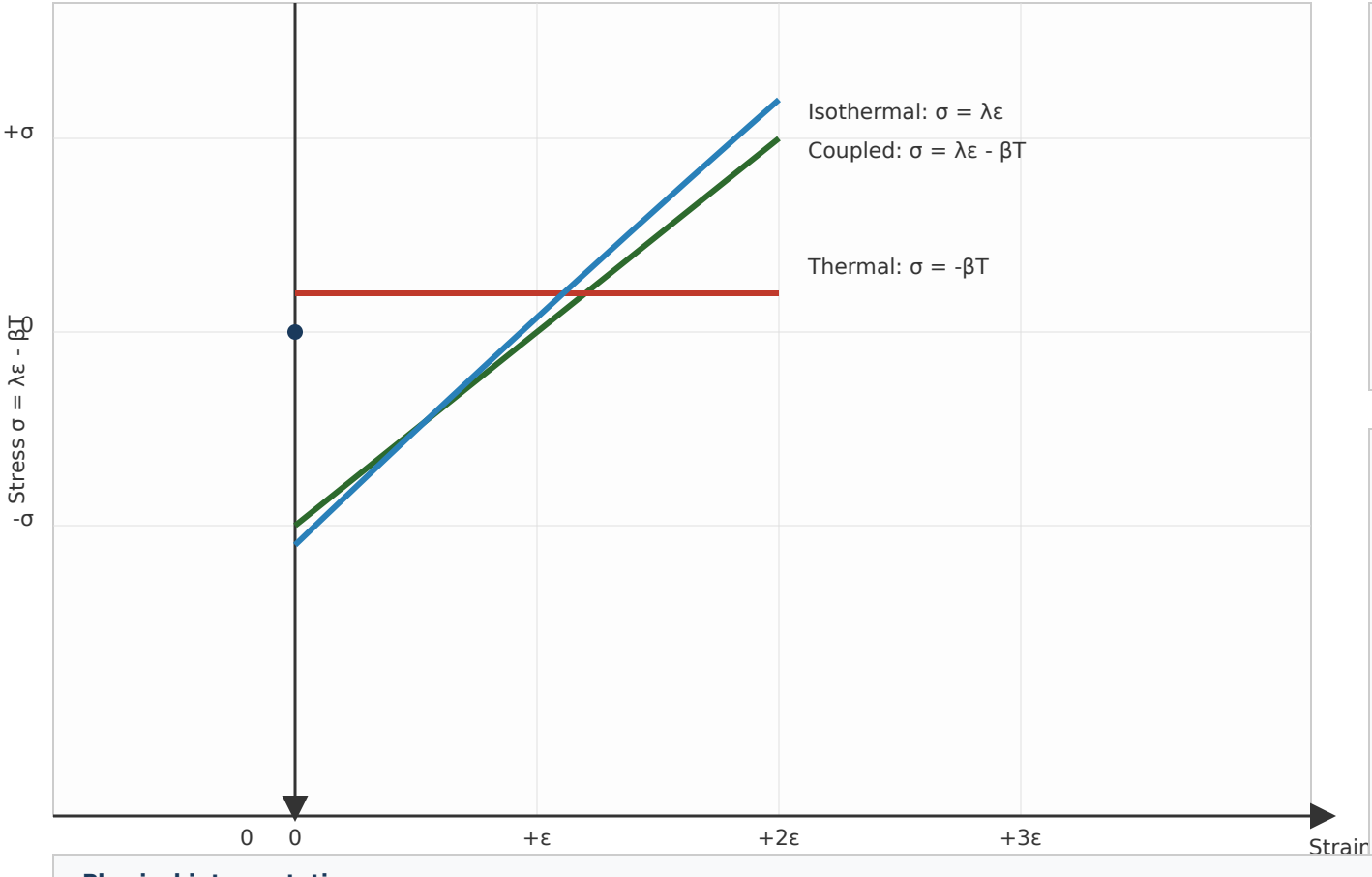
Constitutive relations and thermoelastic coupling

Stress-strain response under isothermal and non-isothermal conditions (Section 3.5)



Constitutive regimes

$\sigma = \lambda\varepsilon\delta + 2\mu\varepsilon$
Hooke's law (elastic)

$\sigma = -3\beta T$
Thermal stress (free expansion)

$\sigma = C\varepsilon - \beta T\delta$
Thermoelastic coupling

$\psi = \psi(\varepsilon, T)$
Free energy (Coleman-Noll)

Key equations (Section 3)

Eq. (3.21) Linear isotropic stress
Eq. (3.22) Entropy-strain coupling
Eq. (3.18) Coleman-Noll restrictions
Eq. (3.29) Energy flux J
Eq. (3.31) Interface conditions

Physical interpretation

The first two terms of Eq. (3.21) are classical elastic stress (Hooke's law).
The last term (-βTδ) is thermal stress: temperature changes induce stress.
Eq. (3.22) shows deformation changes entropy: mechanical work is converted to thermal energy.
This bidirectional coupling is the physical origin of thermoelastic wave propagation.
Generalized models modify different parts of this foundation; each requires model-specific equations and assumptions.